

Phase 2: Requirement Analysis

AI Comic Story Creator — SmartBridge Project Submission

1. Functional Requirements

- The system shall accept a natural-language story premise and target page count as input.
- The system shall generate a structured JSON script (title, art style, characters, pages, panels, dialogue) using Gemini 1.5 Pro.
- The system shall generate a 'Visual Anchor Descriptor' per character (hair, outfit, facial structure) using Gemini 1.5 Flash, enforced across every panel the character appears in.
- The system shall generate per-panel image-rendering prompts (camera angle, lighting, environment, dialogue bubble anchor points) via the Panel Layout Director agent.
- The system shall synthesize panel artwork via Imagen 3 / Imagen Ultra from the combined prompt (art style + character anchor + action + environment).
- The system shall calculate page grid layouts (1-panel splash, 4-panel standard, 3-panel dynamic, etc.) based on panel count.
- The system shall composite dialogue bubbles and captions onto rendered panels.
- The system shall export the finished comic as a print-ready PDF (300 DPI CMYK) and/or CBZ/CBR digital format.

2. Non-Functional Requirements

- **Performance:** panel generation should run asynchronously so the user isn't blocked on long-running Imagen calls.
- **Scalability:** the rendering workers should scale independently from the script-generation workers.
- **Reliability:** transient Imagen API rate-limit errors should be retried automatically via a task queue.
- **Consistency:** character appearance should remain visually stable across all panels of an issue.
- **Cost efficiency:** repeated prompt tokens (character/style descriptors) should be cached rather than resent on every panel call.
- **Portability:** output formats should support both print (PDF/CMYK) and digital reading (CBZ, Webtoon).

3. Inputs & Outputs

Stage	Input	Output
Story Architect	User premise, page count, genre	7-8 page structured script (JSON)
Character Manager	Script JSON, character profiles	Visual anchor descriptors
Panel Director	Script beats + visual anchors	Image prompts + dialogue bubble anchors
Image Renderer	Extended image prompts	High-resolution panel artwork
Typesetter/Publisher	Panel images + bubble metadata	Final PDF / CBZ pages

4. Tools & Technology Requirements

- Google Gemini 1.5 Pro / 2.0 Flash for scripting, visual direction, and character profiling.
- Google Imagen 3 / Imagen Ultra for panel image synthesis.
- Python with the google-genai SDK for orchestration.
- Pillow / Canvas API for bubble drawing and compositing.
- ReportLab for PDF assembly; ImageMagick/Pillow for CMYK conversion.
- FastAPI on Google Cloud Run for the API layer; Google Cloud Tasks or Celery for async job handling; Cloud Storage for renders and exported files.